Logic: switches that decide

IoT · Level 2 · Lesson 2



Today we will

- Describe AND, OR and NOT with switches and truth tables.
- Build AND and OR from two switches, in series and in parallel, with kit or on paper.
- Write a classroom rule (a fan, an alarm, a stair light) as logic.

Three gates

- AND: all inputs on
- OR: any input on
- NOT: flips the input
- A truth table checks every case

Human logic gates

1. Warm-up (5 min): When should the classroom fan run? Collect rules.
2. Teach (10 min): AND, OR and NOT with switches on the board.
3. Act it out (12 min): Pupils are the gates, with 1 and 0 cards.
4. Tables (8 min): Groups write the fan rule as a truth table.
5. Wrap-up (5 min): Share one real rule written as logic.

Prepare 1 and 0 cards in advance; scrap paper is fine. If you have kit, wire the AND and OR circuits before class and test them.

Let’s discuss

1. Where do you see AND and OR rules at school or at home?
2. Why write a truth table before building anything?
3. Which classroom rule would you add NOT to, and why?

Exit ticket

- Write the fan rule with AND, and fill in its four-row truth table.